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Physics-Based Event Attribution From Sparse PMUs With Recovery Memory

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ABSTRACT An electrical network can continue oscillating after a load is restored or a line is reclosed. A later intervention can then resemble the recovery of an earlier one. We address this ambiguity by comparing physical histories from eight sparse PMUs. DAE-derived switched predictors use voltage, oriented terminal current, frequency and ROCOF, start from estimated past states, and preserve recovery through restoration. A ten-second state history supports screened onset proposals; conditional evidence integrates load severity and updates fixed network hypotheses. Active-event and originating-cause masses keep ongoing interventions separate from their remaining response. Completed development streams give 53/54 timely family/source decisions with 4 extra alarms; nominal gains one timely decision at a higher alarm burden. Independent confirmation is in progress. In a separate known-onset load test with supplied load family, conditioned signatures reduce median relative severity error, evaluated at the generating source, from 1.2% to 0.329%, while matching reference-only source accuracy. The contribution is a specified recovery-aware attribution mechanism and an evaluation separating detection, timely location and restoration. The scope is a known load/line catalog; probability calibration, all-family diagnosis and real-time deployment remain unestablished.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [1]; physical outage detectors compare network hypotheses sequentially [2], [3]; and simulated fault trajectories support diagnosis from sparse PMUs [4]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.
2) A comparison that separates detection, correct source

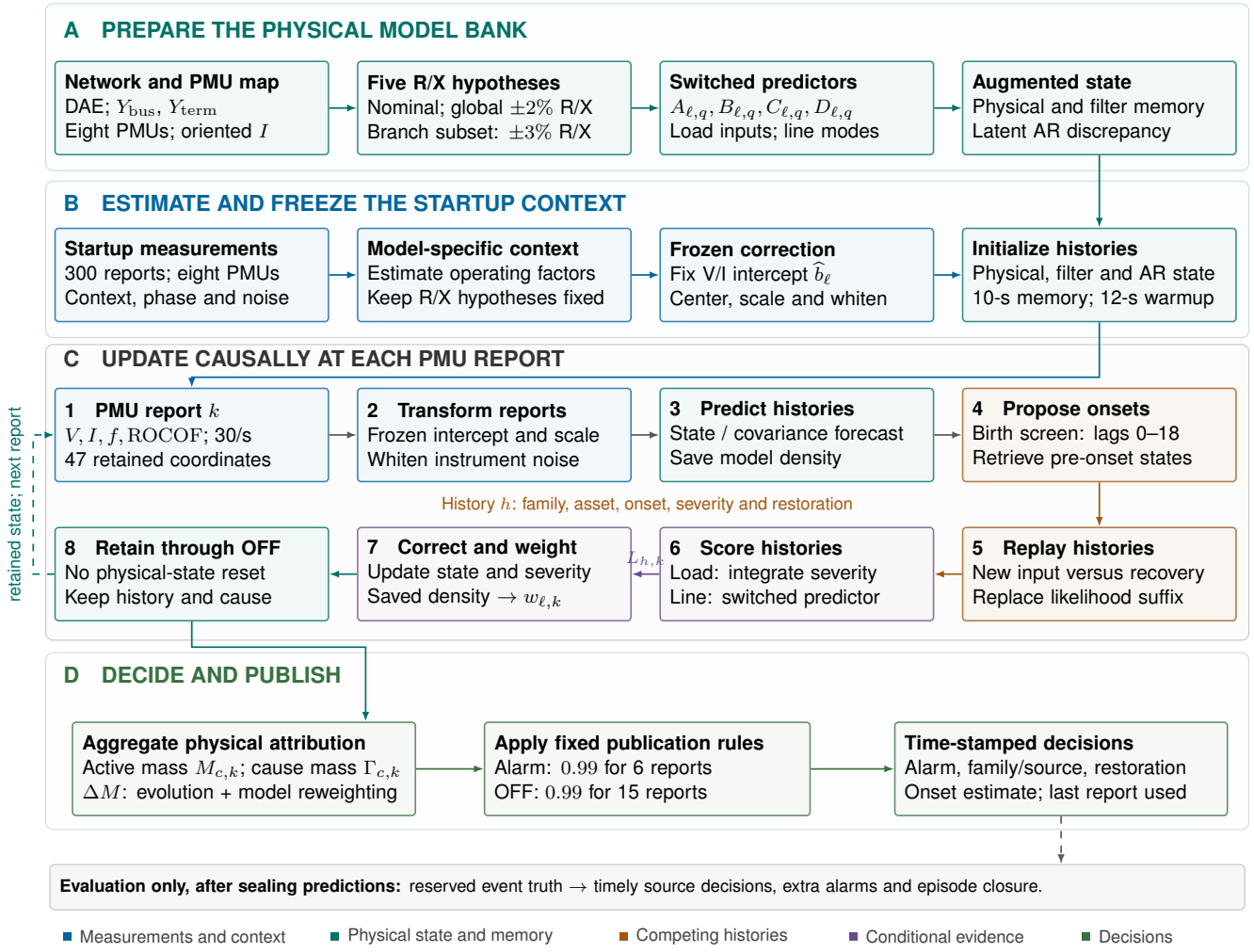


FIGURE 1. Implemented physical-history attribution, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed.

attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.

3) A known-family, known-ON/OFF IEEE-39 component test that separates correction of the pre-event reference from a change in the event signature. The test estimates context from PMUs and measures response fidelity, source decisions and continuous severity.

A further physical control asks whether recovery increases source separation. It distinguishes retaining a consistent prediction from obtaining additional information about the source; it is not a comparison of causal detectors.

Completed development streams expose a trade-off: the five-model reference obtains 53/54 timely family/source decisions with 4 extra alarms, while nominal gains one timely decision at a higher alarm burden. The prespecified confirmatory comparison is still running without method changes. The claim is scoped to the declared load/line catalog. These simulations do not establish an all-family hackathon score, calibrated event probabilities or real-time deployment.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [1], [5], [6]. Functional observability asks whether a requested state quantity can be recovered [7], [8]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [9], [10].

Physical hypotheses support outage inference from phasor angles [11], Bayesian inference with uncertain states [12], and sequential detection under stochastic operation [2], [3].

Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics [13]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [14]–[16].

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [4]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [17]. Multiple-model filtering already propagates conditional state/mode hypotheses [18]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [19]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [20]–[22]. We compare two fixed ExtraTrees adapters from the SGSMA project. Their performance tests transfer of those implementations, not the limits of all supervised learning. A known physical catalog and source-excluded supervised training also impose different information constraints.

Physical stochastic models support latent-state estimation [23], [24], while joint DAE parameter inference addresses uncertain generators and networks [25]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [26]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

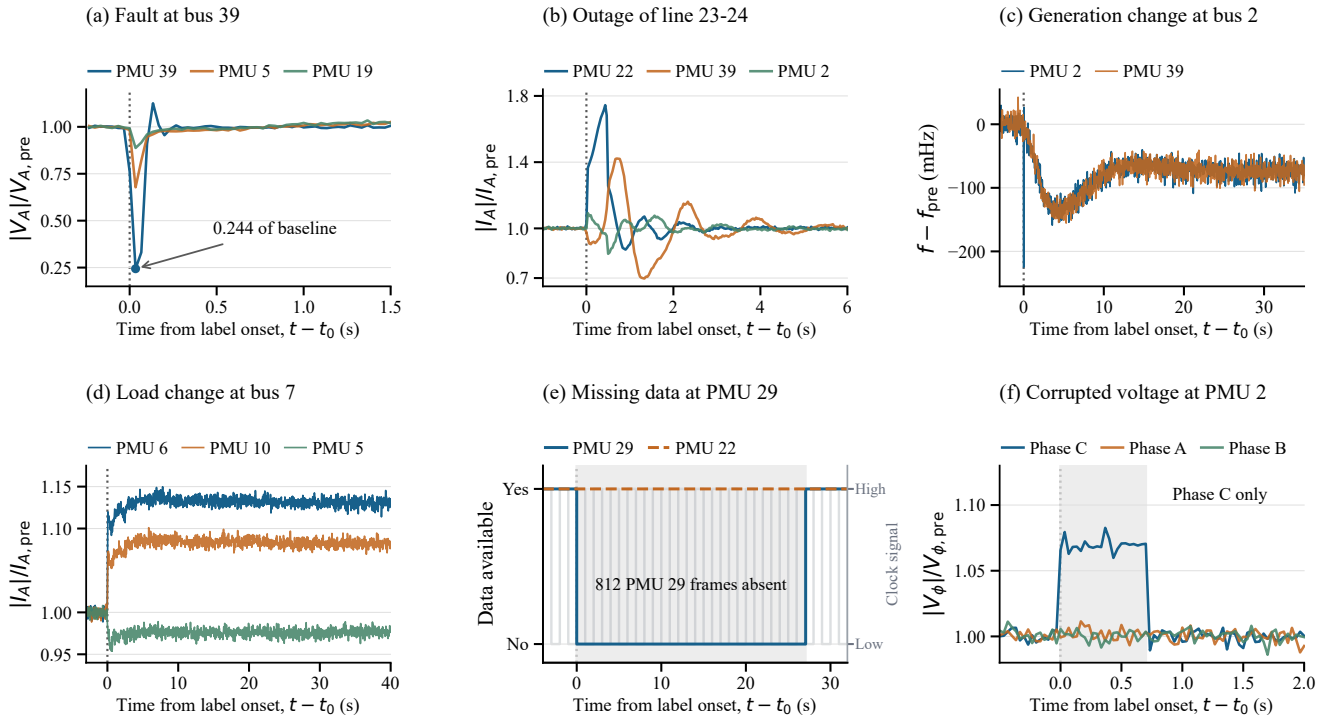


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine, archived as R11, uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

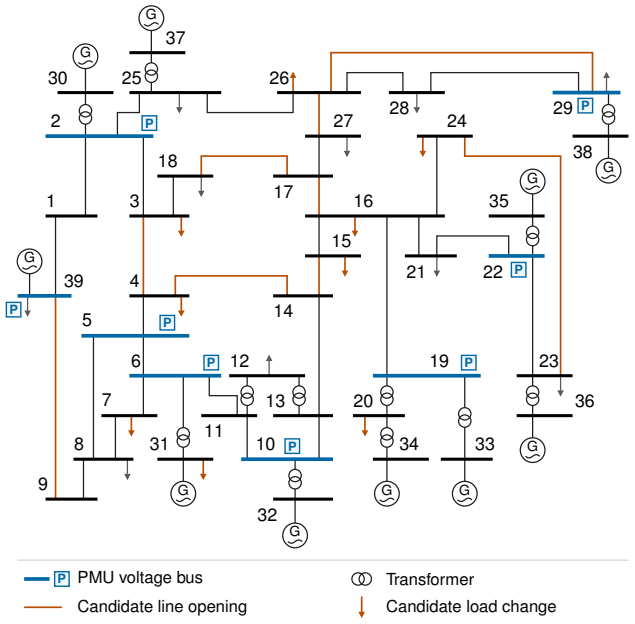


FIGURE 3. IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u), \\ \delta\dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = {} & \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ & + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

The conditional state mean is represented as $m_h + d_h a$. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral above. The regime covariances are obtained from nominal-operator Riccati equations; R11 does not propagate an independent exact covariance for every switched history. At episode closure, it retains the mixture mean of the augmented state and records the omitted between-history spread. Thus state continuity and conditional Bayesian amplitude integration are implemented, while a fully Bayesian posterior over states, context and all histories is not.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{14}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{15}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{16}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\begin{aligned} \Delta M_{c,k} = {} & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \tag{17}$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

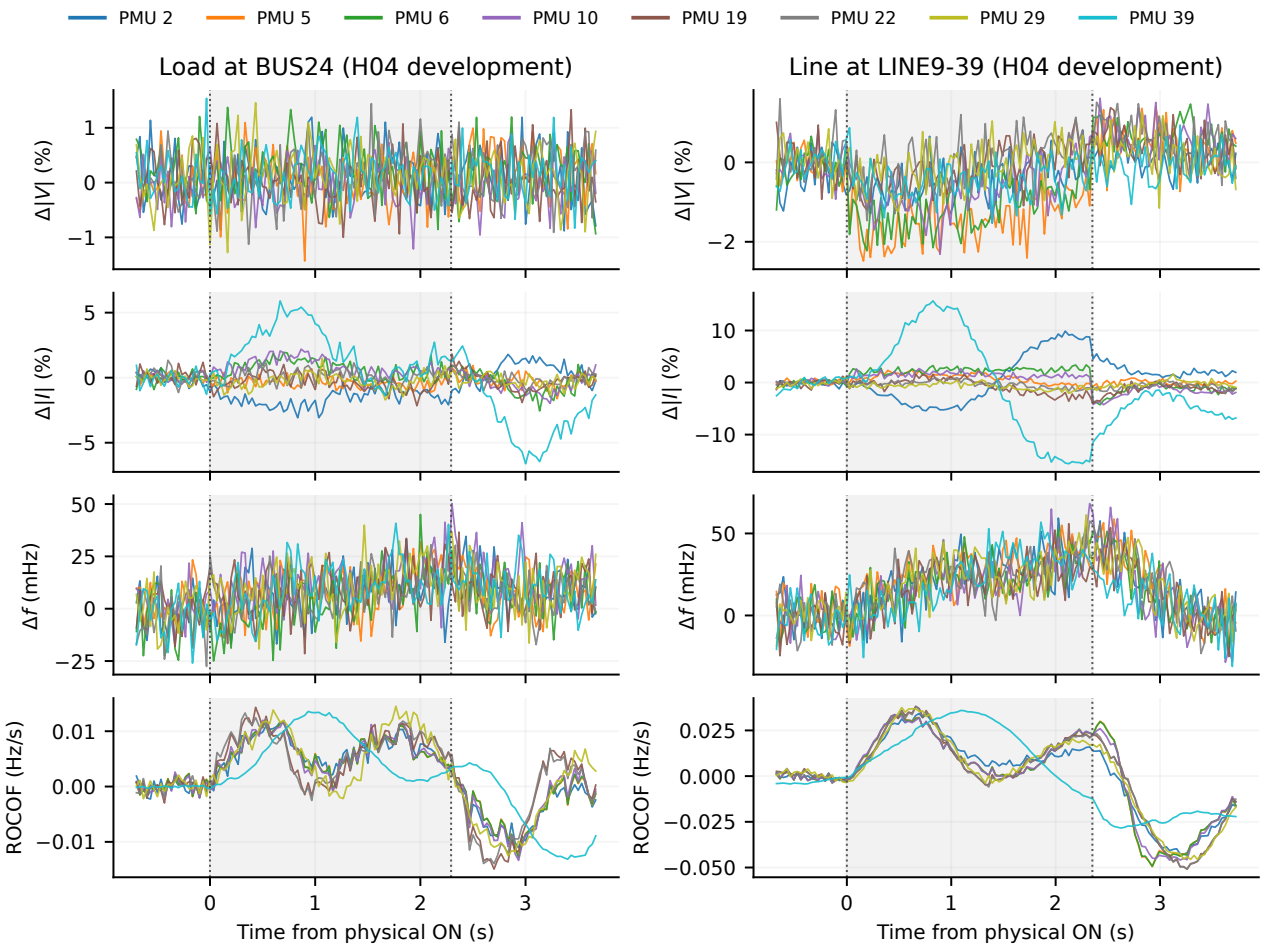


FIGURE 4. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The reproducibility archive maps these operations to the frozen `transitions`, `process`, and `predictive_log_density` routines and preserves the complete publication configuration.

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize condi-

tional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The simulator is the IEEE-39 Julia/PowerDynamics research candidate F01. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on H04 or C01. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [2]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared F01 adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign, C01, therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent evaluation is in progress. The following section reports completed development results only.

TABLE 3. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Signature component test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Recovery geometry control	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; not R11 performance

VI. CONTINUOUS-STREAM RESULTS

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 4). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 6 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (17) separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation.

TABLE 4. What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

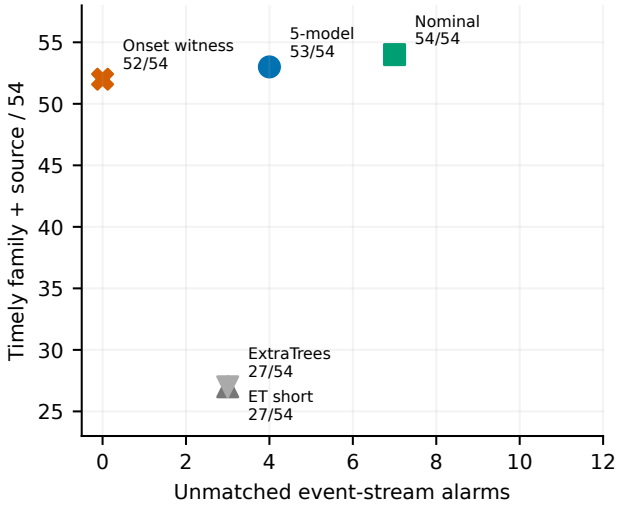


FIGURE 5. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

This negative experiment leaves R11 and nominal frozen for C01.

VII. OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of R11.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{18}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{19}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 8 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

VIII. INTERPRETATION AND LIMITS

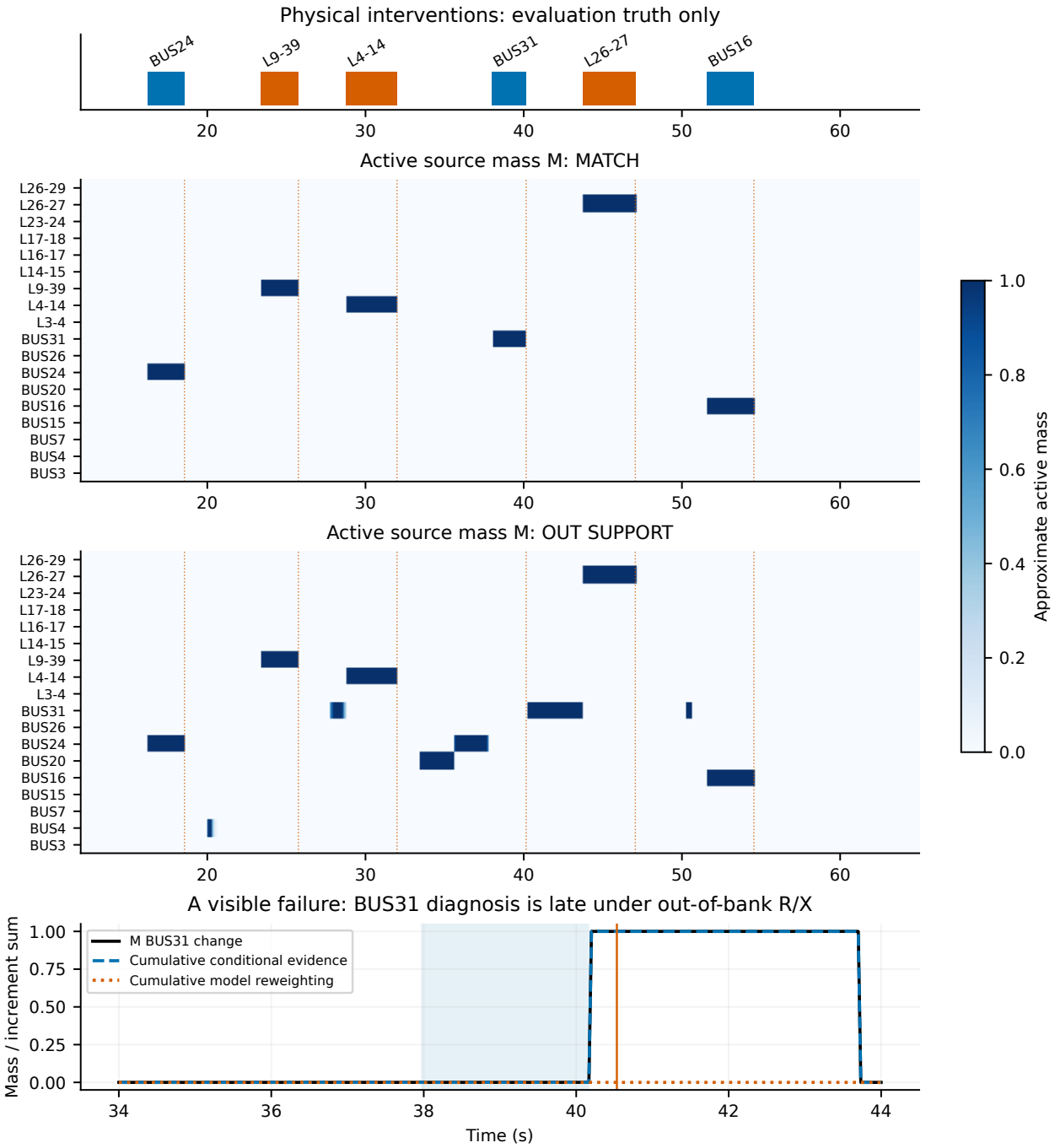


FIGURE 6. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

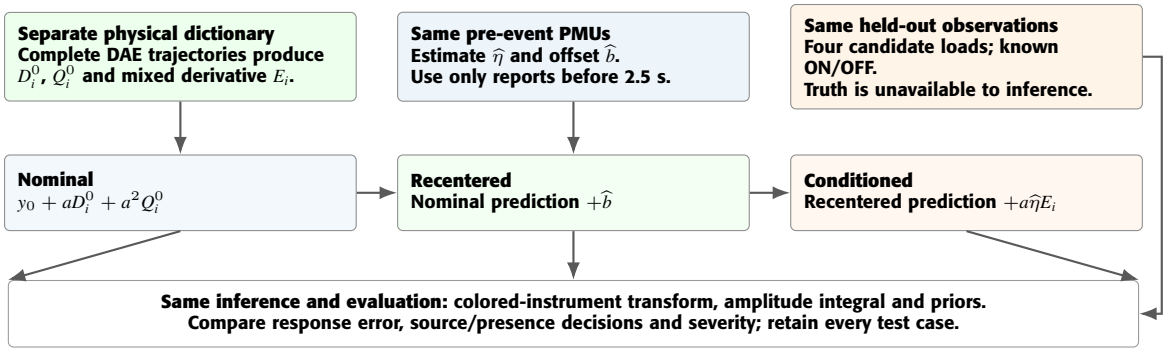


FIGURE 7. Executed signature ablation, separate from the stream algorithm in Fig. 1. The three variants share the data and decision calculation; only reference and event-response terms differ.

TABLE 5. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

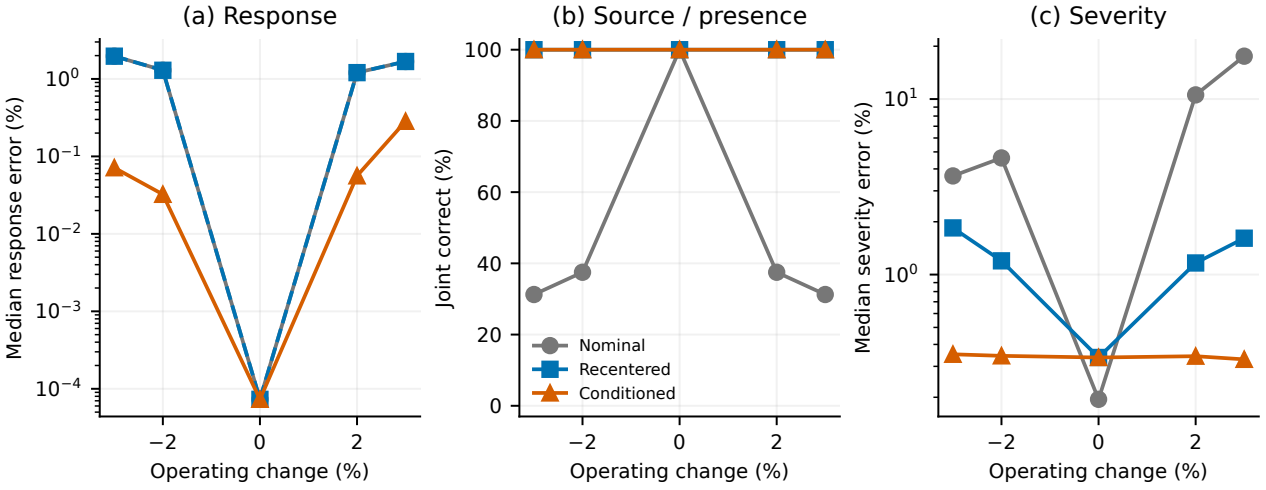


FIGURE 8. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{20}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted

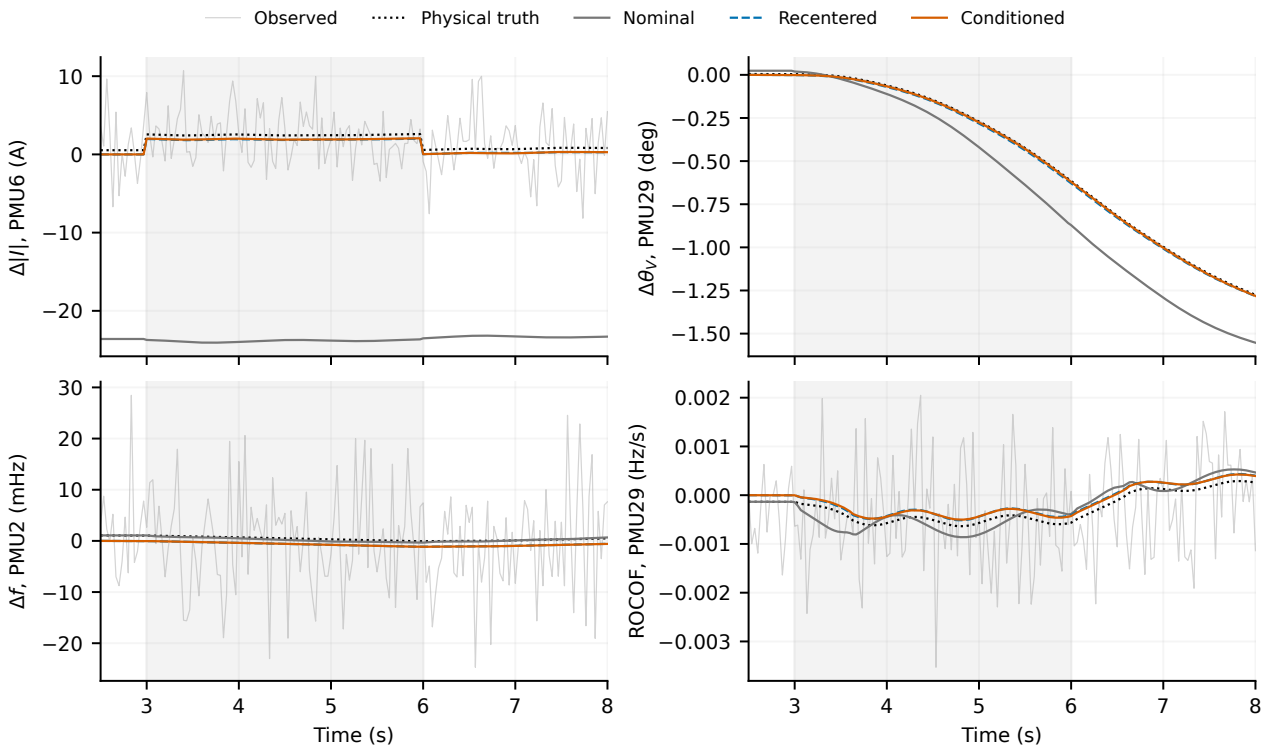


FIGURE 9. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{21}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [9], [10].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The supplementary archive retains the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios

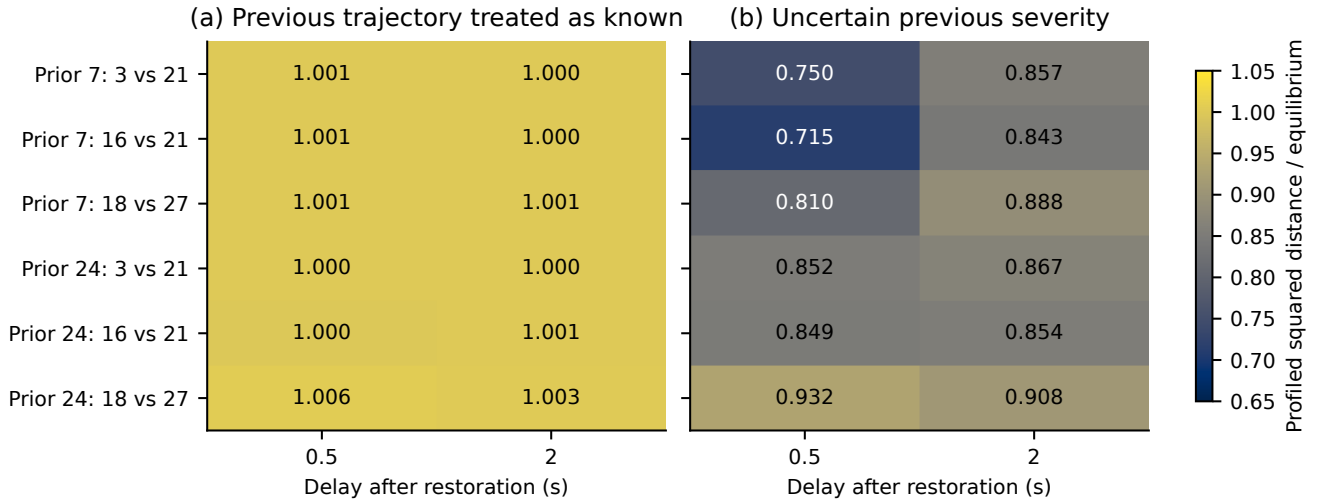


FIGURE 10. **Full-F01 physical control, not an R11 performance comparison. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

exceeds one (Fig. 10). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an R11 improvement nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

C. OPERATIONAL MEANING AND SCOPE

The physical distinction underlying the stream method is simple: restoration changes an input or topology, but does not erase the machine state. Retaining that state lets recovery compete with a new intervention. The development comparison also shows why alarm count and timely location must be reported together. Stricter admission can suppress extras while rejecting a real load. The existing admission ablation does not isolate the contribution of every recovery-memory operation.

The signature experiment addresses a different source of error. Its corrected variants receive the same pre-event measurements and offset. Any difference between them therefore comes from the event–context term and its estimated context, rather than a privileged baseline or longer window. This isolates the tested response approximation, but not an improvement of the unchanged causal detector. Known timing, one scalar context and four load sources make it a controlled component test.

Our construction builds on trajectory-based diagnosis, sequential likelihoods and multiple-model filtering. Their general principles are not new contributions here. The methodological claim concerns the specified competition among intervention and recovery histories; the experiments measure its operational behavior. Numerical superiority is claimed only where a common-data comparison supports it, not over Warner or other procedures that were not executed.

The continuous catalog contains nine load buses and nine lines and requires an intervention-free startup. Missing PMUs, unknown families, source-excluded causal transfer and field operation are untested. F01 retains a provisional BUS5 current terminal and excludes BUS39 current angle. Agreement with this research model does not establish the producer's CT placement or exact reproduction of RAW0001. Local predictors, history pruning, covariance approximations and fitted startup parameters also limit probability claims.

F01's noise port applies temporal and cross-PMU structure within each signal family. Its archived cross-family factor covariance is not applied, and frequency/ROCOF noise is added after phase estimation. The inference comparison therefore concerns this declared simulator, not the complete dependence structure of a physical PMU installation.

The two ExtraTrees adapters are particular frozen implementations. Their load-transfer failure does not rule out learned load diagnosis or establish a better official hackathon score. Chen's eligible test concerns persistent line openings; its F01 adapter has DC/white-injection assumptions that do not capture all instrumental effects. Its mismatches and censored decisions remain part of the comparison.

Joint context uncertainty, unrestricted simultaneous sources and calibrated posterior probabilities remain unvalidated. Such extensions must preserve source-discriminating response directions and account for dependence between

estimation error and measurement noise.

Finally, processing reports causally is not proof of real-time throughput. Observation time, computation and queue delay must be measured separately. The present implementations remain research inference services rather than validated protection or control systems.

IX. CONCLUSION

We formulated sparse-PMU event attribution through competing physical histories. Load/line predictors start from estimated past states and retain the cause of the response after restoration. Active mass M and its increment ΔM expose the changing explanation rather than replacing the physical dynamics.

In completed development streams, the mixture obtains 53/54 timely family/source decisions with 4 extra alarms. Nominal gains one timely decision at a higher alarm burden. Independent confirmation remains in progress.

The controlled load test distinguishes reference correction from signature conditioning. The mixed term reduces response error and median relative severity error, but does not establish a directional improvement in joint source decisions. This component result does not change the frozen causal method.

The result is a specified physical attribution procedure and a testable separation of recovery, reference error and source discrimination. Its scope remains the declared simulation contracts. It does not establish all-family diagnosis, universal source recovery, calibrated probabilities or real-time deployment.

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